

Natural Computing SoSe25

Exercise Sheet 6 — July 10th, 2025

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1 Orcas and Coconuts

For the following tasks we define an artificial chemistry $\mathcal{C} = (\mathcal{X}, \mathcal{R}, A)$ where $\mathcal{X} = \{\text{COCONUT}, \text{TREE}, \text{ORCA}, \text{HORSE}, \text{SHORE}\}$, $\mathcal{R} = \{R_1, \dots, R_7\}$ with

$$\begin{aligned}R_1 &= \text{TREE} \rightarrow \text{TREE} + \text{COCONUT}, \\R_2 &= \text{COCONUT} + \text{SHORE} \rightarrow \text{TREE} + \text{SHORE}, \\R_3 &= \text{COCONUT} + \text{COCONUT} \rightarrow \text{HORSE}, \\R_4 &= \text{ORCA} + \text{COCONUT} \rightarrow \text{ORCA}, \\R_5 &= \text{ORCA} + \text{ORCA} \rightarrow \text{ORCA}, \\R_6 &= \text{ORCA} + \text{HORSE} \rightarrow \text{ORCA}, \\R_7 &= \text{ORCA} + \text{HORSE} + \text{HORSE} \rightarrow \text{HORSE},\end{aligned}$$

and $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

(a) **Give a possible evolution of the soup**

$$X_0 = \{\text{COCONUT}, \text{SHORE}, \text{ORCA}\}$$

for as many time steps as necessary until a HORSE appears. (Hint: Instead of random choices, you can decide which rules to apply to make the process quicker. Use the table below.)

time step	chosen rule	current soup
0	none	$\{\text{COCONUT}, \text{SHORE}, \text{ORCA}\}$
1		
2		
3		
4		
5		

(b) Let $X = \{\text{ORCA}, \text{ORCA}, \text{HORSE}, \text{HORSE}\}$. **What is $\mathcal{A}(X, \mathcal{R})$? What are all possible final soups if we keep running reactions on X for as long as we can?**

(c) Now let $X = X_{\text{Orcas}} \uplus \{x\}$ where X_{Orcas} is a multi-set consisting of 100 ORCA particles and $x \in \mathcal{X}$ is one single mystery particle. We now run studies on the soup X for an extremely long time and every time we do so, we observe that in late soups there are no ORCA particles anymore. **What is the only plausible mystery particle x ? State your reasoning.**

2 DNA-Soup

In the lecture we will discuss an artificial chemistry $(\mathcal{X}, \mathcal{R}, A)$ for a DNA-inspired soup as shown below. We assume that the bindings between DNA strands are labeled with elements from $\{A, C, G, T\}$.¹ In our case we have also defined $-A = T$ and $-C = G$ as A binds with T and C binds with G.

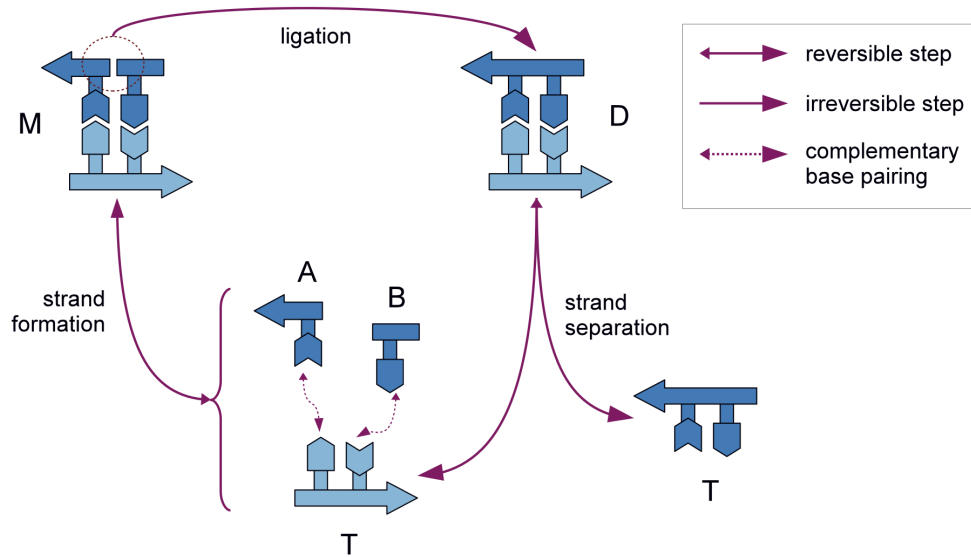


Figure 1: A model of the minimal self-replicator demonstrated by Von Kiedrowski (*A self-replicating hexadeoxynucleotide*. *Angewandte Chemie International Edition in English* 25.10, 1986). Taken from *Artificial Chemistries* (Banzhaf, Yamamoto. MIT Press, 2015), Ch. 6.2, Fig. 6.6

Let $\mathcal{X} = \{A, C, G, T\}^{2 \times 2} \cup \{A, C, G, T\}^2 \cup \{A, C, G, T\}^1$.

¹These names are naturally inspired by the real-life DNA base pairs guanine–cytosine and adenine–thymine. Also see <https://en.wikipedia.org/wiki/DNA>.

Let $\mathcal{R} = \{R_{1,-,-}, R_{2,-,-}, R_{3,-,-}, R_{4,-}\}$ with

$$\begin{aligned}
R_{1,X,Y} &= (X \ Y) + (-X) + (-Y) & \xrightarrow{0.5} & \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} \\
R_{2,X,Y} &= \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} & \xrightarrow{\quad} & (X \ Y) + (-Y \ -X) \\
R_{3,X,Y} &= (X) + (Y) & \xrightleftharpoons[0.01]{0.01} & (X \ Y) \\
R_{4,X} &= \emptyset & \xrightarrow{0.1} & (X)
\end{aligned}$$

for all $X, Y \in \{A, C, G, T\}$.
Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

(a) **Give an evolution of the soup**

$$X_0 = \{C, (T \ G)\}$$

for all reaction steps indicated in the table. (In our case we are always ‘lucky’ enough for our intended reaction to succeed; hence you may ignore the chance p in any reaction with $\xrightarrow{p}$ for now.)

time step	chosen rule	current soup
0	none	$\{C, (T \ G)\}$
1	$R_{4,A}$	
2	$R_{3,A,C}^{\rightarrow}$	
3	$R_{3,T,G}^{\leftarrow}$	
4	$R_{1,A,C}$	
5	$R_{2,A,C}$	

(b) After working with the above soup encoding, a colleague proposed a slightly altered definition for the state space $\mathcal{X}'$. “Instead of using different types for single, smaller, or combined molecules, it might be more helpful to encode DNA molecules as fixed-size matrices that can contain empty fields.”, he said. Give an adjusted ruleset $\mathcal{R}'$, so that the reaction of task (a) still functionally works the same, but accomodates the state space

$$\mathcal{X}' = \{\text{A, C, G, T, 0}\}^{2 \times 2}.$$

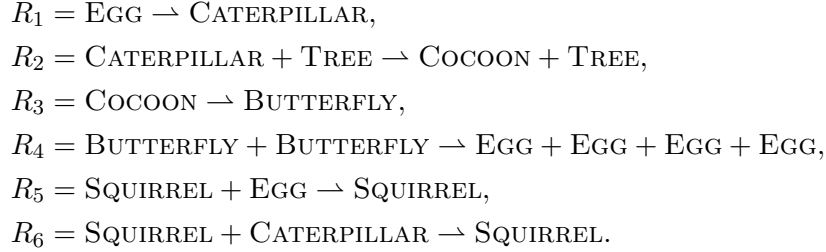
(c) Give X'_0 , i.e., the counterpart to our initial soup

$$X_0 = \{\text{C, (T G)}\}$$

encoded according to our altered state-space definition $\mathcal{X}'$.

3 Squirrels and Butterflies

For the following tasks we define an artificial chemistry $\mathcal{B} = (\mathcal{X}, \mathcal{R}, A)$ where $\mathcal{X} = \{\text{EGG}, \text{CATERPILLAR}, \text{COCOON}, \text{BUTTERFLY}, \text{TREE}, \text{SQUIRREL}\}$ and $\mathcal{R} = \{R_1, \dots, R_6\}$ where



For the reactor function A we assume that it first picks a uniform random rule $R \in \mathcal{R}$ and then checks if all particles necessary to apply R exist within the current soup X_t at time t . If such particles exist, A picks a random set of suitable particles X' to match the left hand side of rule R and then returns (X', R) . Whenever suitable particles for R do not exist within X_t , A again picks a uniform random rule $R \in \mathcal{R}$ and tries to use that.

(a) **Give a possible evolution of the soup**

$$X_0 = \{\text{CATERPILLAR}, \text{BUTTERFLY}, \text{TREE}\}$$

for as many time steps as necessary until an EGG appears. (Hint: Instead of random choices, you can decide which rules to apply to make the process quicker. Use the table below.)

time step	chosen rule	current soup
0	none	$\{\text{CATERPILLAR}, \text{BUTTERFLY}, \text{TREE}\}$
1		
2		
3		
4		
5		

(b) **Give initial populations $X_0^{stay}, X_0^{grow}, X_0^{shrink}$ so that when we evolve these populations separately, i.e., we compute $X_t^{stay}, X_t^{grow}, X_t^{shrink}$, respectively, for increasingly large values of t by applying the artificial chemistry $\mathcal{B}$, it holds that $|X_t^{stay}|$ remains constant for all t , $|X_t^{grow}|$ eventually increases with sufficiently large t , and $|X_t^{shrink}|$ eventually shrinks with sufficiently large t .**

(c) For this task, consider a soup of artificial chemistry $\mathcal{B}$ with the initial population

$$X_0 = \{\text{EGG}, \text{SQUIRREL}, \text{TREE}\}.$$

What are the chances that anywhere in the future evolution of X_0 a BUTTERFLY appears? Show your reasoning. You can assume that all executions of A eventually terminate. You can draw all possible paths for the evolution of X_0 to derive the chances. Hint: Review exactly which behavior A uses.

(d) For this task, consider another soup of artificial chemistry $\mathcal{B}$ with the initial population

$$X_0 = \{\text{EGG}, \text{SQUIRREL}, \text{SQUIRREL}, \text{TREE}\},$$

i.e., this time starting with two SQUIRRELS instead of only one. **Does this addition of another SQUIRREL affect the chances of a BUTTERFLY appearing, which you computed for task (c)? Give the chance and show your reasoning.**

Hint: Review exactly which behavior A uses.